

OC IT Projects — Chapter 8

IP Addressing — Full Review + Complete Quiz

English version — includes all 25 quiz questions, key points, mistakes, and answer key.

1. Key Points

- IPv4: 32 bits = 4 octets; each octet = 8 bits.
- Binary values: 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1.
- CIDR: /24 = 24 Network bits + 8 Host bits.
- Host bits: 32 – CIDR.
- Total IPs: $2^{(\text{Host bits})}$.
- Usable Hosts: Total IPs – 2 (Network + Broadcast).
- Block Size: 256 – the changing subnet-mask octet.
- Network: first address of the block. Broadcast: last address of the block. Usable hosts are between them.
- VLSM: Users + 2 → next power of 2 → determine CIDR. Training example: 54→/26, 32→/26, 5→/29, 3→/29.
- IPv6: hexadecimal groups separated by colons; consecutive zero groups can be compressed with ::.

2. Quick Subnetting Table

CIDR	Host	Total	Usable	Mask	Block
/24	8	256	254	255.255.255.0	256
/25	7	128	126	255.255.255.128	128
/26	6	64	62	255.255.255.192	64
/27	5	32	30	255.255.255.224	32
/28	4	16	14	255.255.255.240	16
/29	3	8	6	255.255.255.248	8
/30	2	4	2	255.255.255.252	4

3. Your Mistakes to Review

- /26 → 6 host bits → $2^6 = 64$ total → 62 usable.
- /27 → 5 host bits → 32 total → 30 usable; block size 32.
- /28 → 4 host bits → 16 total → 14 usable; block size 16.
- Do not confuse Network with first usable host: for /27 second subnet, Network=.32 and first usable=.33.
- Broadcast is the last address of the block. Example: /26 subnet .192-.255 has last usable .254 and Broadcast .255.
- 30 users → $30+2=32$ → /27.
- 54 users → $54+2=56$ → next power of 2 = 64 → /26.
- 5 users → $5+2=7$ → next power of 2 = 8 → /29.
- Training VLSM allocation: 54→/26, 32→/26, 5→/29, 3→/29.

4. Complete Chapter 8 Quiz — All 25 Questions

Q1. How many bits are in an IPv4 address?

- A) 16
- B) 64
- C) 32
- D) 128

Q2. How many bits are in each IPv4 octet?

- A) 8
- B) 4
- C) 16
- D) 32

Q3. What is 11000000 in decimal?

- A) 168
- B) 192
- C) 224
- D) 128

Q4. What is 25 in 8-bit binary?

- A) 00010101
- B) 00011001
- C) 00011100
- D) 00110001

Q5. What does /24 mean?

- A) 24 Host bits and 8 Network bits
- B) 24 Network bits and 8 Host bits
- C) 24 total bits
- D) 24 usable IP addresses

Q6. What is the subnet mask for /27?

- A) 255.255.255.192
- B) 255.255.255.240
- C) 255.255.255.224
- D) 255.255.255.128

Q7. A /26 subnet has how many total IP addresses?

- A) 32
- B) 128
- C) 62
- D) 64

Q8. How many usable host addresses are available in a /26 subnet?

- A) 64
- B) 62
- C) 60
- D) 63

Q9. What is the block size of 255.255.255.224?

- A) 16
- B) 64
- C) 32
- D) 128

Q10. For 192.168.1.0/27, what is the first subnet's broadcast address?

- A) 192.168.1.31
- B) 192.168.1.32
- C) 192.168.1.30

D) 192.168.1.255

Q11. For 192.168.1.0/27, which range represents the second subnet?

- A) Network .31 → Broadcast .62
- B) Network .32 → Broadcast .63
- C) Network .33 → Broadcast .64
- D) Network .64 → Broadcast .95

Q12. For 192.168.1.0/26, what is the third subnet's network address?

- A) 192.168.1.96
- B) 192.168.1.64
- C) 192.168.1.128
- D) 192.168.1.192

Q13. For a /28, how many host bits remain?

- A) 4
- B) 8
- C) 6
- D) 28

Q14. How many total IP addresses are in a /28?

- A) 8
- B) 14
- C) 32
- D) 16

Q15. How many usable hosts are available in a /28?

- A) 16
- B) 14
- C) 12
- D) 8

Q16. You need a subnet for 50 users. Following the training's method of adding 2 addresses for Network and Broadcast, which subnet size is appropriate?

- A) 32 addresses
- B) 64 addresses
- C) 128 addresses
- D) 16 addresses

Q17. A department has 30 users. You add 2 for Network/Broadcast: $30+2=32$. Which CIDR provides exactly 32 total addresses?

- A) /26
- B) /27
- C) /28
- D) /29

Q18. In the training's VLSM example, a department with 54 users is assigned:

- A) /24
- B) /28
- C) /29
- D) /26

Q19. In that same example, a department with 5 users is assigned:

- A) /29
- B) /28
- C) /30
- D) /27

Q20. You have 10.10.10.0/24 and need subnets for 54, 32, 5, and 3 users. Which allocation matches the training?

- A) /27, /27, /29, /29

B) /26, /26, /28, /28

C) /26, /26, /29, /29

D) /25, /26, /29, /30

Q21. What notation is used in IPv6 addresses?

A) Decimal octets separated by dots

B) Hexadecimal groups separated by colons

C) Binary groups separated by slashes

D) Decimal groups separated by hyphens

Q22. What does :: represent when used in IPv6 notation?

A) A subnet mask

B) A network address

C) Consecutive groups of zeros

D) A broadcast address

Q23. Which is a valid compression based on the training example?

A) 2018:0:0:0000:0:000:4815:54ae → 2018::4815:54ae

B) 2018:0:0:0000:0:000:4815:54ae → 2018:0:4815:54ae

C) 2018:0:0:0000:0:000:4815:54ae → 2018::0:4815:54ae

D) 2018:0:0:0000:0:000:4815:54ae → 2018:4815::54ae

Q24. Which statement is correct?

A) /27 has 7 host bits

B) /27 has 5 host bits

C) /27 has 27 host bits

D) /27 has 32 host bits

Q25. You are given 192.168.10.0/28. You calculate: Host bits=4, Total=16, Usable=14, Block=16. Which is the correct first subnet?

A) Network .1, Usable .2-.14, Broadcast .15

B) Network .0, Usable .1-.15, Broadcast .16

C) Network .0, Usable .1-.14, Broadcast .15

D) Network .16, Usable .17-.30, Broadcast .31

5. Answer Key — Complete Quiz

1-C | 2-A | 3-B | 4-B | 5-B

6-C | 7-D | 8-B | 9-C | 10-A

11-B | 12-C | 13-A | 14-D | 15-B

16-B | 17-B | 18-D | 19-A | 20-C

21-B | 22-C | 23-A | 24-B | 25-C

6. Mini Practice

1) 172.16.1.0/27 → Total? Usable? Mask? Block Size?

2) 192.168.20.0/28 → First subnet: Network / Usable / Broadcast?

3) 192.168.20.0/26 → Third subnet: Network / Broadcast?

4) Department needs 14 users → minimum subnet size and CIDR?

5) Department needs 60 users → minimum subnet size and CIDR?

7. Final Memory Card

- Host bits = 32 – CIDR
- Total IPs = $2^{\text{(Host bits)}}$
- Usable Hosts = Total – 2
- Block Size = 256 – mask octet
- Network = beginning of block
- Broadcast = last address of block
- Usable = between Network and Broadcast
- /26 → 64 total / 62 usable / .192 mask / block 64
- /27 → 32 total / 30 usable / .224 mask / block 32
- /28 → 16 total / 14 usable / .240 mask / block 16